

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

Annexure A

Scenario-Based

Code of Conduct:

I **KONGARA JOGESH(192525035)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K.Jogesh

Annexure A

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

- School of Computing → /25
- School of Electronics → /26
- School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

- Determine the network address and broadcast address for each department.
- Identify the valid host IP address range for each subnet.
- Calculate the number of usable host addresses available in each subnet.
- Determine the remaining unused IP address range within the given network block.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block 10.10.0.0/16. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

- Software Development → /24
- Quality Assurance → /25
- Human Resources → /26
- Executive Management → /27

As the network administrator, perform the following tasks:

- Determine the network address and broadcast address for each department.
- Identify the valid range of host IP addresses.
- Calculate the number of usable hosts in each subnet.
- Identify the unused IP address range remaining within the allocated IP block.

3. A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network: 2001:0db8:abcd:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

- Convert the given IPv6 address into its compressed notation.
- Expand the compressed address back into its full hexadecimal representation.
- Identify the network prefix using the /64 prefix length.
- Calculate the total number of host addresses supported within the subnet.
- The total number of possible host addresses available within this network.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

- 192.168.1.0/24
- 192.168.2.0/24
- 192.168.3.0/24

A packet arrives at the router with the destination IP address: 192.168.2.45

As the routing engineer, determine:

- The destination subnet to which the packet belongs.
- The matching routing table entry.
- The next-hop network/interface used to forward the packet.
- The default route that should be used if the destination does not match any routing table entry.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

- Administration LAN → 192.168.10.0/26
- Finance LAN → 192.168.10.64/26

The router interfaces are configured with:

- Administration Gateway → 192.168.10.1
- Finance Gateway → 192.168.10.65

As the network administrator, perform the following tasks:

- Determine the network address and broadcast address for each LAN.
- Identify the valid host IP address range.

ANSWER 1: Smart Campus Subnetting (192.168.100.0/24)

Given block: 192.168.100.0/24 = 256 total addresses. Method: VLSM (Variable Length Subnet Masking), largest subnet allocated first.

Step 1: Identify subnet sizes required

- /25 $\rightarrow 2^{(32-25)} = 128$ addresses (School of Computing)
- /26 $\rightarrow 2^{(32-26)} = 64$ addresses (School of Electronics)
- /27 $\rightarrow 2^{(32-27)} = 32$ addresses (School of Mechanical Engineering)
- Order of allocation (largest $\rightarrow$ smallest): Computing, Electronics, Mechanical.

Step 2: Allocate School of Computing (/25)

- Range occupied: 192.168.100.0 – 192.168.100.127
- Network Address = 192.168.100.0
- Broadcast Address = 192.168.100.127
- Valid Host Range = 192.168.100.1 – 192.168.100.126
- Usable Hosts = $2^7 - 2 = 126$

Step 3: Allocate School of Electronics (/26)

- Starts immediately after Computing block, at 192.168.100.128
- Network Address = 192.168.100.128
- Broadcast Address = 192.168.100.191
- Valid Host Range = 192.168.100.129 – 192.168.100.190
- Usable Hosts = $2^6 - 2 = 62$

Step 4: Allocate School of Mechanical Engineering (/27)

- Starts immediately after Electronics block, at 192.168.100.192
- Network Address = 192.168.100.192
- Broadcast Address = 192.168.100.223
- Valid Host Range = 192.168.100.193 – 192.168.100.222
- Usable Hosts = $2^5 - 2 = 30$

Step 5: Calculate remaining unused range

- Total used = $128 + 64 + 32 = 224$ addresses
- Remaining = $256 - 224 = 32$ addresses
- Unused Range = 192.168.100.224 – 192.168.100.255 (a free /27 block, reserved for future use)

Summary Table

Department	Network Addr.	Broadcast Addr.	Host Range	Usable Hosts
Computing (/25)	192.168.100.0	192.168.100.127	.1 – .126	126
Electronics (/26)	192.168.100.128	192.168.100.191	.129 – .190	62
Mechanical (/27)	192.168.100.192	192.168.100.223	.193 – .222	30
Unused (Free /27)	192.168.100.224	192.168.100.255	.225 – .254	30 (reserved)

ANSWER 2: Multinational Company Subnetting (10.10.0.0/16)

Given block: 10.10.0.0/16 = 65,536 total addresses. Method: VLSM, largest subnet first.

Step 1: Identify subnet sizes required

- /24 → 256 addresses (Software Development)
- /25 → 128 addresses (Quality Assurance)
- /26 → 64 addresses (Human Resources)
- /27 → 32 addresses (Executive Management)
- Order of allocation: Software Development, QA, HR, Executive Management.

Step 2: Allocate Software Development (/24)

- Range: 10.10.0.0 – 10.10.0.255
- Network Address = 10.10.0.0
- Broadcast Address = 10.10.0.255
- Valid Host Range = 10.10.0.1 – 10.10.0.254
- Usable Hosts = $2^8 - 2 = 254$

Step 3: Allocate Quality Assurance (/25)

- Starts right after Software Development, at 10.10.1.0
- Network Address = 10.10.1.0
- Broadcast Address = 10.10.1.127
- Valid Host Range = 10.10.1.1 – 10.10.1.126
- Usable Hosts = $2^7 - 2 = 126$

Step 4: Allocate Human Resources (/26)

- Starts right after QA, at 10.10.1.128
- Network Address = 10.10.1.128
- Broadcast Address = 10.10.1.191
- Valid Host Range = 10.10.1.129 – 10.10.1.190
- Usable Hosts = $2^6 - 2 = 62$

Step 5: Allocate Executive Management (/27)

- Starts right after HR, at 10.10.1.192
- Network Address = 10.10.1.192
- Broadcast Address = 10.10.1.223
- Valid Host Range = 10.10.1.193 – 10.10.1.222
- Usable Hosts = $2^5 - 2 = 30$

Step 6: Calculate remaining unused range

- Total used = $256 + 128 + 64 + 32 = 480$ addresses
- Remaining = $65,536 - 480 = 65,056$ addresses
- Unused Range = 10.10.1.224 – 10.10.255.255 (reserved for future departments/branches)

Summary Table

Department	Network Addr.	Broadcast Addr.	Host Range	Usable Hosts
Software Dev. (/24)	10.10.0.0	10.10.0.255	10.10.0.1 – .254	254
Quality Assurance (/25)	10.10.1.0	10.10.1.127	10.10.1.1 – .126	126
Human Resources (/26)	10.10.1.128	10.10.1.191	10.10.1.129 – .190	62
Executive Mgmt. (/27)	10.10.1.192	10.10.1.223	10.10.1.193 – .222	30

ANSWER 3: IPv6 Migration for Smart Hospital

Given: 2001:0db8:abcd:0000:0000:0000:0000/64 — 128-bit address, 8 groups of 16 bits each.

Step 1: Apply compression rules

- Rule 1: Omit leading zeros within each group.
- Rule 2: Replace one contiguous run of all-zero groups with '::' (used only once).
- Compressed Notation = 2001:db8:abcd::/64

Step 2: Expand back to full form

- Restore all 8 groups to 4 hex digits each, replacing '::' with the correct number of zero groups.
- Expanded Address = 2001:0db8:abcd:0000:0000:0000:0000:0000

Step 3: Identify the network prefix (/64)

- First 64 bits (first 4 groups) = network/subnet portion.
- Last 64 bits (last 4 groups) = interface identifier (host portion).
- Network Prefix = 2001:0db8:abcd:0000::/64 (compressed: 2001:db8:abcd::/64)

Step 4: Calculate total host addresses in the subnet

- Host bits available = $128 - 64 = 64$ bits
- Total addresses = $2^{64} = 18,446,744,073,709,551,616$

Step 5: Determine usable host addresses

- IPv6 has no broadcast address (multicast is used instead).
- Nearly the entire 2^{64} space ($\approx 1.8 \times 10^{19}$ addresses) is usable for host interfaces.
- This confirms /64 comfortably supports thousands of connected medical devices and IoT sensors without exhaustion risk.

ANSWER 4: ISP Routing Table Lookup

Routing table: 192.168.1.0/24, 192.168.2.0/24, 192.168.3.0/24. Destination packet: 192.168.2.45.

Step 1: Apply the subnet mask to the destination address

- Mask /24 = 255.255.255.0
- 192.168.2.45 AND 255.255.255.0 = 192.168.2.0
- Destination subnet = 192.168.2.0/24

Step 2: Perform longest-prefix match against the routing table

- 192.168.1.0/24 → No match
- 192.168.2.0/24 → Match (selected)
- 192.168.3.0/24 → No match

Step 3: Determine the next-hop interface

- 192.168.2.0/24 is a directly connected network.
- Next hop = the router's own interface on 192.168.2.0/24 (e.g., 192.168.2.1).
- Packet is forwarded directly onto that LAN and delivered to 192.168.2.45 via ARP resolution.

Step 4: State the default-route fallback rule

- If no entry matched, the router would use the default route 0.0.0.0/0.
- This typically points to the ISP's upstream gateway.
- If no default route exists and no entry matches, the packet is dropped and an ICMP 'Destination Unreachable' message is returned to the sender.

Result Table

Item	Value
Destination Subnet	192.168.2.0/24
Matching Routing Entry	192.168.2.0/24
Next-Hop Interface	Router interface on 192.168.2.0/24 (e.g., 192.168.2.1)
Default Route (no match)	0.0.0.0/0

ANSWER 5: Two-Department LAN via Layer-3 Router

Given: Administration LAN = 192.168.10.0/26 (gateway 192.168.10.1); Finance LAN = 192.168.10.64/26 (gateway 192.168.10.65). Each /26 = $2^{(32-26)} = 64$ addresses.

Step 1: Determine network and broadcast addresses

- Administration LAN (192.168.10.0/26): Network = 192.168.10.0, Broadcast = 192.168.10.63
- Finance LAN (192.168.10.64/26): Network = 192.168.10.64, Broadcast = 192.168.10.127

Step 2: Determine valid host ranges

- Administration LAN: 192.168.10.1 – 192.168.10.62 → 62 usable hosts ($2^6 - 2$)
- Finance LAN: 192.168.10.65 – 192.168.10.126 → 62 usable hosts ($2^6 - 2$)

Step 3: Assign sample host addresses

- Router gateway occupies the first usable address in each subnet.
- Administration LAN — PC-1: 192.168.10.2, PC-2: 192.168.10.3, PC-3: 192.168.10.4
- Finance LAN — PC-1: 192.168.10.66, PC-2: 192.168.10.67, PC-3: 192.168.10.68

Step 4: Assign default gateway for each LAN

- Administration LAN → Default Gateway = 192.168.10.1
- Finance LAN → Default Gateway = 192.168.10.65
- Every host must point to its own subnet's router interface for inter-LAN and external routing.

Summary Table

LAN	Network	Broadcast	Host Range	Gateway
Administration	192.168.10.0	192.168.10.63	.1 – .62	192.168.10.1
Finance	192.168.10.64	192.168.10.127	.65 – .126	192.168.10.65

Rubrics for Calculation — Scenario Based Assessment

Criteria	Wt. %	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Clear, complete understanding of scenario/context	Good understanding, minor gaps	Partial understanding; misses key aspects	Misinterprets/fails to understand
Identification of Key Issues	20%	Accurately identifies all relevant issues/factors	Identifies most key issues, minor omissions	Limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies concepts effectively to the scenario	Applies concepts with minor errors	Limited/partially correct application	Unable to apply relevant concepts
Decision Making	20%	Well-reasoned, logical decisions with strong rationale	Reasonable decisions with some justification	Weak decisions, limited justification	No clear decisions/rationale
Clarity of Response	15%	Clear, structured, logically presented	Generally clear, minor issues	Somewhat unclear/poorly structured	Disorganized, difficult to understand